

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final
Duration: 1 hour 50 min

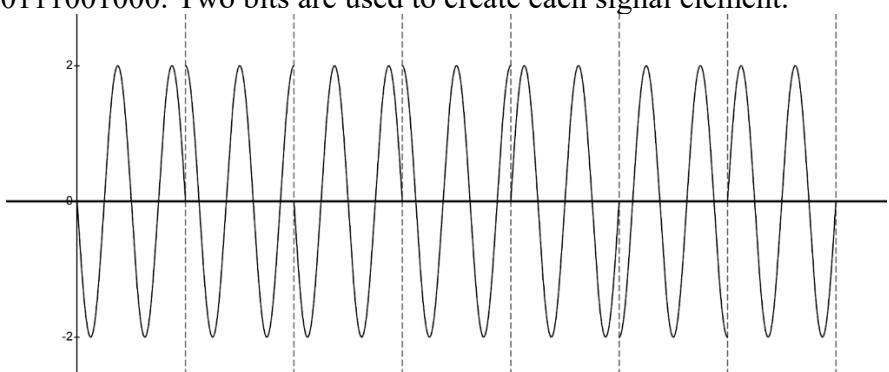
Semester: Summer 2025
Full Marks: 60

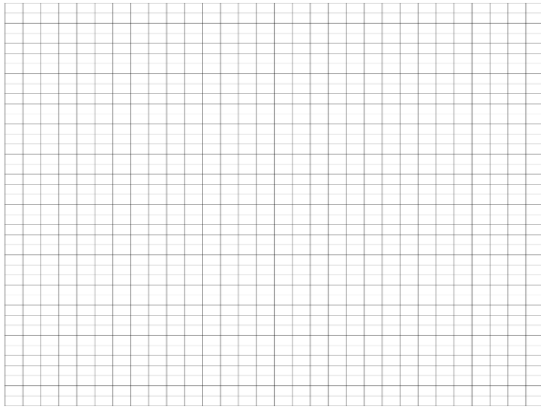
CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:
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1. [CO2]	a)	In multilevel FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10v and phase is 180 degrees. If the frequency changes according to the following table, draw the modulated signal for the bit sequence 1100010110. <table border="1" style="margin-left: auto; margin-right: auto;"><thead><tr><th>Bits</th><th>No. of cycles of the signal elements</th></tr></thead><tbody><tr><td>00</td><td>2</td></tr><tr><td>01</td><td>1</td></tr><tr><td>10</td><td>4</td></tr><tr><td>11</td><td>3</td></tr></tbody></table> 	Bits	No. of cycles of the signal elements	00	2	01	1	10	4	11	3	[5]
	Bits	No. of cycles of the signal elements											
00	2												
01	1												
10	4												
11	3												
b)	The following analog signal has been received for the bit stream 01110111001000. Two bits are used to create each signal element. 	[1+4]											

		- Which type of shift keying is used here? Draw the constellation diagram for the above shift keying technique. 	
2. [CO3]	a)	FOUR analog channels are to be multiplexed using FDM. Required frequency bandwidths for each channel are 30 kHz, 35 kHz, 45 kHz, and 60 kHz respectively. A guard band is required between every two adjacent channels. The size of each guard band is 40% of the difference between the bandwidths of those two channels. Find the total bandwidth required in the output channel. Show the FDM scheme in frequency domain representation assuming the starting frequency to be 200 kHz.	[4+2]
	b)	- Mention some demerits of Synchronous TDM technique. - Consider a high-performance computing system with 100 devices, but only 40% devices are active at any given time. Which TDM technique should be used for this system, and why?	[2+2]
	c)	A system is using Time Division Multiplexing (TDM) to combine 3 channels together. Channel 1 is transmitting 15 kb (kilobits) in 3 seconds, channel 2 is transmitting 10 kb in 2 seconds and channel 3 is transmitting 15 kb in 1 second. If 5 bits from each channel and extra 3 synchronization bits are used to create one frame, Answer the following questions: - How many input channels are being used for final TDM? - What is the frame size - What is the frame rate? - What is the output data rate?	[8]

3. [CO4]	a)	Suppose you are given the following CRC Codeword 1010001 , where the first 4 bits are the original Dataword . - How many 0's are augmented at the sender side? - You are given two possible divisors: 1101 and 1011. Determine which one is the correct divisor.	[1+4+1]										
	b)	Observe the Block Coding scheme below. <table border="1"><thead><tr><th>Dataword</th><th>Codeword</th></tr></thead><tbody><tr><td>00</td><td>0011001</td></tr><tr><td>01</td><td>0100101</td></tr><tr><td>10</td><td>1010011</td></tr><tr><td>11</td><td>?</td></tr></tbody></table> - If XORing the second and third codeword produces the fourth codeword, find the fourth codeword. Calculate the total number of codewords which are invalid. - How many errors can be detected and corrected in the above scheme? - Suppose the codeword received is 0111001. Is there any error in this received codeword? If there is indeed an error, find the original Dataword. Show necessary calculation.	Dataword	Codeword	00	0011001	01	0100101	10	1010011	11	?	[1+1+4+3]
	Dataword	Codeword											
00	0011001												
01	0100101												
10	1010011												
11	?												
c)	Assume a packet is made only of four 8-bit words (CD) ₁₆ , (1F) ₁₆ , (4A) ₁₆ , and (B0) ₁₆ . Show the checksum at the sender. - If the second data item is changed to (1E)₁₆ and the fourth data item is changed to (B1)₁₆ during transmission, check if the receiver can detect any error in this case. Explain the reasons of the receiver's error detection state in (ii)	[2+2+1]											
4. [CO4]	a)	In a warehouse, autonomous delivery robots send status messages to a central monitoring system using the Pure ALOHA protocol. Robot R1 begins transmitting a message at t₁ = 10 ms and the transmission takes 3.6 ms. The first bit of the message reaches the monitoring system at 10.6 ms. Determine the vulnerable period for R1 along with the time window. Justify whether a collision will occur if another robot, R2, starts transmitting at t₂ = 14 ms.	[3]										

	b) A shared bus channel uses 1-persistent CSMA. The bus length is 2.5 km; signal propagation speed is 2×10^8 m/s and the bandwidth of the channel is 9 Mbps. Calculate the propagation delay across the channel. Calculate the vulnerable time for this scenario. - A station senses the channel idle and begins transmitting a 1500-bits frame at $t = 25$ ms. Determine when another station located at the farthest end of the bus will receive the last bit of the frame. - If the network uses p-persistent CSMA instead, explain whether the possibility of collision reduces or not. - In CSMA/CD, what happens if a station detects a collision?	[2+3 +2+2]
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